

Q2 2023 Earnings Call

Company Participants

- Farhan Ahmad, Head of Investor Relations
- Manish Bhatia, Executive Vice President, Global Operations
- Mark Murphy, Executive Vice President and Chief Financial Officer
- Sanjay Mehrotra, President and Chief Executive Officer
- Sumit Sadana, Executive Vice President and Chief Business Officer

Other Participants

- Ambrish Srivastava, Analyst, BMO Capital Markets
- Brian Chin, Analyst, Stifel
- Harsh Kumar, Analyst, Piper Sandler
- Karl Ackerman, Analyst, BNP
- Mehdi Hosseini, Analyst, SIG
- Sidney Ho, Analyst, Deutsche Bank
- Srinij Pajjuri, Analyst, Raymond James
- Vijay Rakesh, Analyst, Mizuho
- Vivek Arya, Analyst, Bank of America

Presentation

Operator

Thank you for standing by, and welcome to Micron's Post Earnings Analyst Call. At this time, all participants are in a listen-only mode. After the speakers' remarks, there will be a question-and-answer session. (Operator Instructions)

I'd now like to introduce your host for today's program, Farhan Ahmad, Vice President, Investor Relations. Please go ahead, sir.

Farhan Ahmad {BIO 20748275 <GO>}

Thank you, and welcome to Micron Technology's fiscal second quarter 2023 sell-side call back. On the call with me today are our Chief Business Officer, Sumit Sadana; our EVP of Global Operations, Manish Bhatia; and our CFO, Mark Murphy.

As a reminder, the matters we are discussing today include forward-looking statements regarding market demand and supply, our expected risk, results and other matters. These forward-looking statements are subject to risks and uncertainties that may cause actual results to differ materially from the statements made today. We refer you to the documents we filed with the SEC, including our most recent Form 10-K and 10-Q, for a discussion of risks that may affect our future results.

I'll now turn the call over to the operator to begin the Q&A session.

Questions And Answers

Operator

Certainly. (Operator Instructions) And our first question comes from the line of Vivek Arya from Bank of America. Your question, please.

Q - Vivek Arya {BIO 6781604 <GO>}

Thank you for the call back. I had a two-part question. One is, how far is Micron in terms of your cash cost in DRAM and NAND? And what do you think is the delta for your competitors, right now?

And then maybe one for Mark. Mark, even when I take your inventory write-down expectations for May, it still seems like the ending inventory would be \$7 billion, \$7.5 billion, that would still be higher than what you had exiting the last fiscal year. Is that a comfortable range to be in or do you think that still leaves open the room for more inventory write-downs in the future?

A - Mark Murphy {BIO 16136048 <GO>}

Let me start, maybe, and then go to the cost question. So, yeah, the question -- yeah, we're not comfortable with where our inventory levels are. And separate and apart from the write-down question, Vivek, look, we do have elevated inventory levels. We expect pre-write-down, the DIOs have peaked in the second quarter, and we would expect over time for the supply/demand balance to improve on customer inventories improving and volumes increasing sequentially and eventually working inventory levels down. But it's going to take some time and, yeah, we would like -- yeah, certainly targets have not changed around 100 to 110 days of inventories on-hand, again going to take some time.

Yeah. The question of write-downs is a function of the quantity of inventories and the cost, of course, but also pricing. So that's why we made the point on the call that -- and indicate that, yeah, write-down assumptions are sensitive to those factors including price. And if prices and our outlook are better than expected, then, yeah, we'll have, yeah, less write-downs of inventory, potentially even less than the third quarter. If inventory -- our price projections worsen versus our current view, we could have larger write-downs, for example, in the third quarter and maybe in the fourth quarter. It just depends on the shape of the recovery. And we have a certain view, which is incorporated in the write-offs that we took.

A - Manish Bhatia {BIO 18045807 <GO>}

Okay. Vivek, and then on your first question about cash costs, I can take. This is Manish. So, we've said that we are not yet at cash costs for either DRAM and NAND, so really the underutilization actions that we've taken have really been around managing the overall supply/demand environment for managing our inventory and trying to manage our supply to come down in line with demand that we see.

And with regard to competitors, I can't really comment other than to say -- and we've given some disclosures in the past about how technology leadership, 1-alpha and 176 last year has definitely helped us to become very competitive in costs. So, we feel good about where we are and where our cash cost position would be relative to others as well.

Q - Vivek Arya {BIO 6781604 <GO>}

And if I could quickly follow up. Mark, any -- I know, it's never a simple question, but modeling your gross margins past Q3 is quite challenging. Any rough guidelines you could give to us so that we at least have some kind of boundary conditions on how to model gross margins for the next several quarters?

A - Mark Murphy {[BIO 16136048 <GO>](#)}

It's a good question, Vivek. And anticipating the question, I did in the other call make an effort to provide a profile on gross margin. And, of course, it's -- as we just discussed, it's difficult because at these levels of profitability and the gross margins will be very sensitive to pricing assumptions and cost actions such as underutilization.

So, with that qualification, we did say -- if you look at our reported margins, we would expect the 2Q to be the trough. And then on lower inventory charges in third quarter, we expect a sequential improvement. And then in the fourth quarter, as mentioned, based on our current pricing assumptions, we would have a small or no write-off. And then, of course, as the underutilization charges, they weigh on the fourth quarter and first quarter, and then improve gross margins then improved through the year, that's not reported.

Now, if you strip out just the write-downs in the second quarter and third quarter, the second quarter would be 7.3% gross margin, the third quarter would be negative 7.5%. So, clearly, that profile is down. And then -- and that's just a function of the pricing environment and, again, the cost of under underutilization, which includes period costs.

Now with this view of pulling those charges out -- those write-downs out, under this view, we would trough in the second quarter and then we would begin to improve off these low levels as we get volume leverage on our period costs as the underutilization effects play through. And then, most importantly, as customer inventories begin to improve and inventories begin to come down and the supply/demand picture begins to improve. So, clearly, the supply/demand balance and supply coming out of the system in concert with volumes increasing sequentially is an important part of the recovery here.

Q - Vivek Arya {[BIO 6781604 <GO>](#)}

Thank you.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

Just wanted to give a quick clarification on the pricing comment as well. As Manish said, the pricing is not close to cash cost in the last couple of quarters, Q2, Q1, but in the guidance that we provided in Q3 with the pricing that we've assumed there and the pricing that -- trends we are seeing in the market, particularly in NAND, definitely close to cash cost for Q3 and getting closer in terms of DRAM as well. But just wanted to highlight that cash cost is obviously different from variable cost. The variable cost is much lower than cash cost, because certain cash costs are fixed in nature. So, wanted to just provide that clarification.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Vijay Rakesh from Mizuho. Your question, please.

Q - Vijay Rakesh {[BIO 5884146 <GO>](#)}

Yeah. Hi, just a quick question, just going back on the inventory side. As you look out exiting May quarter given the incremental \$500 million of write-downs and I know you mentioned 25% wafer start cuts now, and the CapEx tweaks I guess. Would you expect inventory exiting the quarter to be closer to that \$7 billion, lower \$7 billion range? And I have a follow-up.

A - Mark Murphy {BIO 16136048 <GO>}

Yeah, on a dollar basis, we would expect inventory levels to remain elevated for some time, but begin to come down through '24. So, but again, it depends on the nature of the recovery and we do expect bit shipment volumes to increase through the year and inventory days certainly to -- on a pre write-off basis, to come down, and then on a dollar basis, eventually begin to come down.

Q - Vijay Rakesh {BIO 5884146 <GO>}

Got it. And then, on the write-downs between the February and the May quarter, just wondering what would be the split between DRAM and NAND, and in DRAM, is the write-down that you've taken more on the DDR4 or DDR5 side since you'd expect some DDR5 demand to probably pick up into the back-half, so might be some benefits from that, but any color there? Thanks.

A - Mark Murphy {BIO 16136048 <GO>}

Yeah, we just report these write-downs at the corporate level and given how we run the business on precedence and other factors, so we don't break it down by technology or by BU.

Operator

Thank you. One moment for our next question. Our next question comes from the line of Sidney Ho from Deutsche Bank. Your question please.

Q - Sidney Ho {BIO 6922415 <GO>}

Thanks for taking the question. So, I'm still trying to figure out the, you undergo utilization charges you are embedding in the fiscal third quarter guidance, trying to bridge the gross margin from Q2 to Q3, excluding that inventory write-down, you talk about maybe down 15 percentage points. Is there any color you can provide just for fiscal Q3? I know it's up about \$900 million for the fiscal -- for the full year.

A - Mark Murphy {BIO 16136048 <GO>}

Yeah, we're just providing the full year, if we look at -- or I just provided the full year, if you could give me a few minutes, maybe go to next question, I can maybe give you a little bit more detail, but let me work that out.

Q - Sidney Ho {BIO 6922415 <GO>}

Okay. The reason I asked that, just to give you a background is because if I look at that down 15 points, let's assume half of that \$900 million is included in fiscal Q3 that's already 12 points of margin difference. So that implies the ASPs not really declined that much, but I'm not sure what I'm missing. But maybe your second question while you look at that, are you still expecting to have production cuts now 25% and 20% through the end of this fiscal year? Are there any plans to go beyond the current fiscal year?

A - Mark Murphy {BIO 16136048 <GO>}

Yeah, let me maybe answer your first question. I've already got the answer on the first one. So, of the \$900 million, not quite half of it is in the third quarter and then, we've got amount split between some second and some fourth. And as it relates to -- we've not commented on utilization beyond '23. Sanjay mentioned on the call that we may. We'll have to evaluate it based on market conditions and so forth.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

That's right.

Q - Sidney Ho {[BIO 6922415 <GO>](#)}

Okay, so the (inaudible) through this yea?

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

And Sidney, one thing I do want to clarify is that the under-utilization charge that Mark mentioned, a portion of it is going to be in the write-down amount as well. So just -- that is the piece that you have to take into account in.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Yeah just to be clear and I think I mentioned this on the main call that, the reason the proportion is higher to the total under-utilization charge is because of that write-down accounting and its effect.

Q - Sidney Ho {[BIO 6922415 <GO>](#)}

Part of that \$500 million of write-downs is this, to see under-utilization charges.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Part of the \$500 would include utilization also, those higher cost. Because utilization -- under-utilization creates higher cost inventories. So by definition, as you're writing that inventory down, part of that charge would be that under-utilization related cost.

Q - Sidney Ho {[BIO 6922415 <GO>](#)}

Got it. Maybe a quick follow-up, if I may. If you exclude the inventory write-downs, can you say why the DRAM gross margin is negative in fiscal Q3. And at what point does it make sense to cut further production or completely stop production, probably not realistic, since you want to maintain your share?

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

Yeah, we are not giving out specific gross margin breakout -- breakdown between DRAM and NAND, particularly with all of these charges that we are talking about related to under-utilization, inventory, write-down et-cetera. So definitely, we'll provide more color when we report results. But having said that, broadly speaking, our goal is to certainly keep flat with share at customers and to not drive the pricing down to gain share, just we are a follower on price, and we try to just keep our bit share flat and manage our business.

A - Manish Bhatia {[BIO 18045807 <GO>](#)}

And Sidney, one more thing that Mark mentioned on the main call was that the gross margins, excluding the impact of the write-downs in Q3 was 7.5% negative. And obviously, the DRAM margins are a lot better than NAND. That is something that we can say while we are not providing you the details and split between the DRAM and NAND gross margins.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

And Sidney, one other thing, on the under-utilization charge, which I mentioned, that's inclusive of costs that are absorbed in the inventories and also period costs. And just keep in mind that, if you're then translating what portion of that goes into the write-down, it's clearly only the portion that's related to the inventory costs. So, just wanted to make sure that distinction was noted.

Operator

Thank you. Our next question comes from the line of Harsh Kumar from Piper Sandler. Your question please.

Q - Harsh Kumar {BIO 3235392 <GO>}

Yeah, hey guys. Thanks for letting me ask the question. I was just curious, very roughly, what is the price difference between DDR4 and DDR5 right now that you guys are seeing? And then also, maybe you could help us think about how much of the DRAM revenues is DDR5 right now and how much do you think is or when do you think the crossover will occur for you guys?

A - Manish Bhatia {BIO 18045807 <GO>}

Yeah, so in terms of DDR4 versus DDR5, we still have healthy premium between DDR4 and DDR5. DDR5 does sell at a premium. The premium is bigger in the data center than it is in the client space. And both data center and client, DDR5 is going to continue ramping through this current and next calendar year and we expect crossover to happen in both segments around mid-calendar 2024 from a bid perspective. The crossover will happen earlier on a revenue basis, but will happen around mid-2024 on a bit basis between the two.

Q - Harsh Kumar {BIO 3235392 <GO>}

Understood. Thank you guys.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Ambrish Srivastava from BMO. Your question please.

Q - Ambrish Srivastava {BIO 4109276 <GO>}

Hi, thank you. Excuse me, Mark. I know you've been very helpful in answering all these questions of the inventory. I had a couple of clarifications. At the beginning of your prepared remarks, you said this is not due to obsolescence, and then did I get it right that in the fiscal third quarter, there will be \$300 million included 3 Included in that from the \$1.4 billion write down?

A - Mark Murphy {BIO 16136048 <GO>}

Yeah, the reference to that \$300 million, Ambrish, was the 1.4 creates lower-cost inventories. And so, the \$300 million was in reference to the benefit associated with that lower-cost inventory sell through. In other words, had we not written it down, we would have \$300 million less income in 3Q versus since we pulled those inventory costs forward into 2Q, we now have \$300 million higher income in 3Q.

Q - Ambrish Srivastava {BIO 4109276 <GO>}

Got it. So, it's not necessarily (multiple speakers)

A - Mark Murphy {[BIO 16136048 <GO>](#)}

That's separate of course, from yes, that the inventory charge in 3Q is the \$500 million. That's an incremental inventory charge.

Q - Ambrish Srivastava {[BIO 4109276 <GO>](#)}

Got it. So the 1.4 and \$500 million targets that you've taken now, we should continue to see benefit of that right if this inventory is not being -- is not going obsolete -- you said there is no-risk of obsolescence then?

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Correct. It's permanently lower-cost inventory.

Q - Ambrish Srivastava {[BIO 4109276 <GO>](#)}

Yeah. Got it.

A - Manish Bhatia {[BIO 18045807 <GO>](#)}

Over the period that we sell.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

Yes and it will move through and pass-on it as per the turns in the business.

Q - Ambrish Srivastava {[BIO 4109276 <GO>](#)}

Got it, and you'll call it out explicitly right every quarter then you get the benefit, correct?

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

I mean, we haven't decided, like we will see how the business proceeds.

Q - Ambrish Srivastava {[BIO 4109276 <GO>](#)}

Okay, and then on-- sorry, just on the minus 7.5 number that you gave for fiscal third-quarter, is that inclusive of the benefit from the \$300 million, because you said trip out everything and just --

A - Mark Murphy {[BIO 16136048 <GO>](#)}

No, I just said that's stripping out the write downs.

Q - Ambrish Srivastava {[BIO 4109276 <GO>](#)}

Okay, got it. And then I had a quick one for Sumit on these CapEx side. Sumit, this is the question I'm sure you get, we all get. What happens to your competitiveness as you push-out your node? What's the right way to think about it in terms of specifically on the DRAM side?

A - Sumit Sadana {[BIO 7009258 <GO>](#)}

So, I'll have Manish answer that question.

A - Manish Bhatia {[BIO 18045807 <GO>](#)}

Sure. So, Ambrish, I think we gave some good color on the call about the progress that we're making on our -- both of our new technology nodes that we announced last year, 1-beta for DRAM and 232-layer for NAND, and feel very good about what those nodes are going to be able to deliver in terms of the per-wafer gain over the prior nodes, both of which were also good node. So feel really good about the intrinsic technology capability.

And then now the yield ramps that we have been able to demonstrate are reaching targeted yields faster than prior nodes really faster than any node in our history. So, we feel very good about our ability to demonstrate technology capability and now what we're preparing to do is to enable those -- both of those technologies across our broad portfolio of products. For example, in 1-beta, we talked about in our prepared remarks, the LP5 part that's going to be generating revenue for us later this year, but we're also going to be utilizing that technology across DDR5 and compute across high-bandwidth memory, graphics, et-cetera.

So we've kind of getting ready-to-use that really strong node across the portfolio. And similarly in NAND, we started in client SSD and we'll move the 232-layer into mobile into datacenter as because we feel very good about that. And as we look-forward to the next-generation nodes for example, 1-gamma making very good progress on 1-gamma. And in particular, that's the node where we're going to introduce EUV and we've actually already taken delivery of, we've mentioned before, of our first production EUV tools in Taiwan.

And we've actually already started using those tools to some degree on limited production in our one alpha and it's already within just a couple of quarters of having landed the tools, demonstrated the ability to match yields with our multi-patterning immersion technology on 1-alpha that's been in production now for a couple of years. So really good progress on EUV, and that gives us confidence that when we introduced 1-gamma, it will also be a strong introduction with good inherent technology capability for the per-wafer gain and cost reduction as well as it's going to wrap predictably and provide a strong cost-reduction for across the portfolio.

Q - Ambrish Srivastava {[BIO 4109276 <GO>](#)}

Okay, thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Karl Ackerman from BNP. Your question please.

Q - Karl Ackerman {[BIO 19693285 <GO>](#)}

Yes, thank you. Two quick ones if I may, just but I was hoping you can discuss your China exposure by product category. My understanding is that you're primarily tied to mobile but if you perhaps rank order the remaining segments will be very helpful as we think about the pending recovery in China.

A - Manish Bhatia {[BIO 18045807 <GO>](#)}

Yeah. I mean, we definitely have good deal business in China, as you said, mobile customers, very strong center in China for mobile OEMs. We also have a very strong automotive business there, as you know, the EUV Vehicle -- sorry, EV vehicle growth drove (multiple speakers) growing really high in China, the penetration is very-high. The uptake is very-high in these cars especially at the medium to higher-end have a lot of electronics, semiconductor components in them, so we

have a good (Technical Difficulty) as you know we are worldwide our market-share leaders in automotive. So, we continue to have a good position there as well.

And then, of course, we have customers on the client side as well as on the data center side, all broad range of customers. So, we are pretty diversified across many different segments and we expect that as the China recovery gains momentum after the reopening, we do expect that there will be some demand uplift coming from China later in calendar '23.

Q - Karl Ackerman {BIO 19693285 <GO>}

That's helpful. Thank you. If I may, my second question, you -- Mark, (inaudible) 2024 CapEx will decline from 2023. Perhaps maybe this is a question for Manish, but is it fair to suggest your implicit outlook is that CapEx will be all brownfield or upgrades and then you would achieve projected bit demand, which should recover in fiscal '24, through the inventory on your balance sheet, unless maybe demand were to recover faster than you currently expect? Maybe just discuss that a bit more color there would be very helpful. Thank you.

A - Mark Murphy {BIO 16136048 <GO>}

Sure, so as we -- we have taken multiple levers to be able to reduce the supply. We reduced CapEx and WFE in particular, as we talked about more than 50% for FY '23 and we've said that for '24, it will be even lower. And then we've also reduced utilization within our facilities. And then, last one is, we've held inventory -- we're holding inventory on our balance sheet.

So as we look-forward into '24, the first thing we'll do is bring our inventories down and that will help us to be able to meet demand and keep up as the market increases, as bit growth increases that we see -- that we talked about occur and our inventories will come down. And then as we start to feel comfortable that we're -- we have a trajectory to reach our inventory targets, of course, we can increase our utilization rates to be able to increase supply as well.

And then the -- CapEx for supply growth would be the kind of the next lever and last lever. So, really right now even in '23, the CapEx that we have, the WFE that we put to work has been primarily -- actually more than primarily, I'd say, explicitly for technology transition and technology learning on these two new nodes that I mentioned and that will be what we will be focused on and even as we go through '24 is bringing the benefit of that new technology to the product portfolio, whether that's 1-beta or whether that's 232-layer.

Q - Karl Ackerman {BIO 19693285 <GO>}

Very clear. Thanks.

Operator

Thank you. One moment for our next question. Our next question comes from the line of Mehdi Hosseini from SIG. Your question, please.

Q - Mehdi Hosseini {BIO 4362002 <GO>}

Yes, thanks for taking my question. A couple of follow-ups. I'm just looking at quarterly CapEx trends throughout several cycles and come up with an estimate of \$4 billion annualized CapEx just for maintenance, just what it would take to Kind of run utilization rate just 70% to 80%. And I think this is obviously making a broad guesstimate, but would -- is the \$4 billion annualized maintenance CapEx in the ballpark?

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Mehdi, I don't know that we've really kind of given a definition. I think, different folks have different definitions about what maintenance CapEx involved. So, I don't think we're really prepared --

Q - Mehdi Hosseini {[BIO 4362002 <GO>](#)}

(multiple speakers) to zero.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Yeah, so if you think about the the CapEx in the way we're -- the elements that we have, the big buckets are of course WFE, which primarily is for technology transitions and when implemented the way we typically have, that does provide bit growth. So the technology transition WFE does provide growth as the primary source of the bit growth that we would target within our supply.

The facilities requirements that we have for implementing those technologies and adding the the new equipment and those tend to be a little bit lumpier and you do those sort of well in advance of when you need them and in fact, I think we mentioned that the mix of our CapEx in the second half of this year is moving more towards -- construction is becoming an increasing part of them in the second half of the year, more than double -- more than twice what it was in the first half of the year. So, sort of starting to prepare clean room space for the future.

And then we have assembly test and other R&D and other areas that we have that are to be able to either provide us with our -- with good cost reductions on our managing our assembly test cost or to provide R&D capabilities for the future. So, these are kind of the big buckets that we typically define our our CapEx.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

Yeah, mainly the CapEx is not really like a consumable. So it's for technology transition, product enablement and advancing to more advanced nodes, which is giving you better cost and better production.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

And more bits, yeah.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

And more bits. So, you shouldn't think that just for maintaining the bit production, we need to invest in CapEx.

Q - Mehdi Hosseini {[BIO 4362002 <GO>](#)}

Got you. Okay. I was just trying to get a sense of where net cash would go to as I try to forecast free cash flows. Okay.

And then just looking beyond the current environment, obviously there's a lot of moving parts with the write-off and underutilization charges. Let's say, February of calendar year '24 Q2 FY '24 and you have your, let's say, some of the DDR4 that are cost down and then DDR5, that is at a premium off of the DDR4. If all equal, even if prices were to go sideways, given this inventory

adjustments if prices would like is stable, you should you should be able to have a better margin profile excluding all the onetime charges.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Yeah, Mehdi, we're not providing specific guidance that far out and we will as we get closer to the date and the market situation becomes clear. We did give you a profile on how to look at the back half of this year and expected improvement on gross margin through '24, assuming that all these factors we laid out move in the direction that we expect. And as you point out, particularly, if you're talking about second quarter '24, some of the write-downs that would occur in third quarter, that lower-cost inventory would be passing through in the second quarter of '24.

Q - Mehdi Hosseini {[BIO 4362002 <GO>](#)}

Can I ask a clarification question? Do you mind if I just have a quick follow-up?

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Go ahead.

Q - Mehdi Hosseini {[BIO 4362002 <GO>](#)}

Thank you. This is a view of next year. As you think about DDR5 premium for both at PC and server applications, would that be -- the premium, would that be also the market pricing for DDR4 or would that be also the cost suggested DDR4 that you have?

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

Yeah. I mean. I think, a lot depends on the market environment and competitive behavior. But generally, obviously, DDR4 as you know also DDR5 tends to be bigger die, more expensive at a module level as well because it has further integration at a module level compared to DDR4. So, it's higher but also higher cost. So. those are some of the factors that coming in to play.

And like you said. I mean, the DDR4, DDR5 transition is going to be taking place, and there is more inventory in DDR4 than there is DDR5. So, lot more DDR5 deployment in the future comes in as new purchases from our customers. That's another thing to look at, but I think broadly speaking, I think you have a number of data points that we have provided to you, which include our bits bottoming in fiscal Q1 of this year are before like write-down in the days of inventory peaking in fiscal Q2, our datacenter revenue bottoming in fiscal Q2, customer inventories improving.

So when you zoom out of all of this, we've also told you that looking at 2025 calendar year, we expect record TAM in that year. So, we do expect that looking out multiple quarters and into '25, that is going to be a very substantial amount of growth that we are expecting in that time-frame in the -- particularly in the late '24 '25 time frame. And obviously, continuing to focus on significant improvements in profitability and trajectory of free cash flow in the interim from here to '24.

Q - Mehdi Hosseini {[BIO 4362002 <GO>](#)}

Got it. Thank you for the details.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

And just one other thing on 1- beta, our but you have to buy DDR5 1-beta technology is optimized for DDR5 in terms of the architecture and the technology. So, we feel very good about what that

will eventually position us for as the market transitions.

Q - Mehdi Hosseini {[BIO 4362002 <GO>](#)}

Yeah. Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Brian Chin from Stifel. Your question please.

Q - Brian Chin {[BIO 5763426 <GO>](#)}

Hi there, good afternoon and thanks for taking the question. Maybe a two-parter on sort of demand. I don't think people who really asked about that, but was the lion's share of the reduction in the counter '23 memory bit demand forecast is that related mainly to data centre? What is your also underlying assumption for datacenter bit demand growth in calendar '23?

And then the second part of that is, does your expectation for a mid-single digit decline in the calendar '23 PC TAM, does that a sellout or a seven-figure? I ask because seven sounds like it could be weaker than than down mid single-digits and I think seven ties, more into kind of activity levels at OEMs.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

Yeah. I mean, this is definitely a good question and we focus on lot of these dynamics with our customers on an ongoing basis, so. in the last three months since our last earnings call, we definitely have been continuously assessing as we always do the level of inventories the customers have and its impact on demand for the near-term planning over the next several quarters.

And so, the degradation in our calendar 2023 outlook for DRAM and NAND came as a result of the assessment of inventories, particularly in the datacenter. And also the pace of progress of inventories, the inventories have been improving. So it's a matter of the rate and pace of that improvement that we have to adjust for in our TAM. But also the extent of the pullback in growth estimates for both PC units as well as smartphones, so smartphones going from low-single digit increase to low-single digit decrease in units in calendar '23 over the last three months and on the PC side, yes, mid-single-digit decrease.

That puts the PC units and this is more of our sell-through comment -- puts the PC unit at a level that is very consistent with the level they were that pre-COVID in 2019 from a global PCs unit sales perspective. So, that's how we think about the PC business and certainly, different customers have different amounts of channel inventory, so their own sell-in versus sell-out rates will be different. And our demand expectations are based on a bottoms-up view of what we are feeling our customers sell-out is going to be, because that's going to sorry -- selling is going to be into the channel that is going to determine the consumption of DRAM and NAND.

Q - Brian Chin {[BIO 5763426 <GO>](#)}

Okay, yeah, okay, great, that makes sense. Maybe just a quick thing in technology kind of separate, but in terms of mix. I know you won't be too specific about percentages and crossover et-cetera, but when next year would you kind of plan for wafer inputs to really began to shift toward 1-beta and 232-layer technologies? I know it may become more observant once you kind of add up some of the utilization charges and et-cetera, right, FIFO inventory et-cetera. But I'm

wondering if there is a kind of a particular point in next year that if you kind of anticipating that to start to pick-up some inertia.

A - Manish Bhatia {BIO 18045807 <GO>}

It's too far-out, Brian, like at this point, they don't think it's worthwhile but like you said, as demand picture improves, we will have more clarity on it. And we just have to see time play-out and as we turn back on utilization and that is currently being implemented. We will also ramp 1-beta at that time when the demand improves.

Q - Brian Chin {BIO 5763426 <GO>}

That's fair enough. Thank you.

Operator

Thank you. One moment for our next question. And our next question from the line of Srini Pajjuri from Raymond James. Your question please.

Q - Srini Pajjuri {BIO 5862807 <GO>}

Thank you. Mark, I'm just trying to figure out the cash-flow for next quarter. Obviously, you've got some help from working capital in the February quarter. So, if you could maybe help us if you're expecting somewhat similar benefit from working capital in the next quarter or two, I think that'll be really helpful.

A - Mark Murphy {BIO 16136048 <GO>}

Yeah. I think, listen on free-cash, I think it's helpful, but maybe step-back and it has been such a sharp and sudden downturn. We know all the factors and we've been aggressively and with even greater intensity through the fall taking actions, CapEx reductions, cutting operating spend, lowering utilization, and even heard some additional actions announced today.

So, we've also over the past several months used our balance sheet to bolster liquidity and make sure that we're making the best long-term decisions for the business and so that's sort of gives some backdrop as to where we are. Our actions are helping improve the financial picture from what it would be otherwise, but it's clear that we need the broader market to recover in more specifically, more supply to come out-of-the system.

But on free-cash flow, we do expect free-cash flow to improve from here, slightly at first, going second to third-quarter as reductions in CapEx are partially offset by weak operating cash flows. We have low volumes and challenging pricing, so that's weighing on operating cash-flow as the benefits of the receivables have waned and the inventories remain elevated. Over-time though, we see free-cash flow improving sequentially by sustaining our capital discipline and on, importantly, improving operating fundamentals.

Specifically, we do see shipments continuing to increase from here as we've talked about customers replenish inventories, inventories began to decline and across the industry and channel supply-demand balance to improve and pricing to revert to more sustainable levels. It's not sustainable where it is. So, between our actions and a healthier industry, we see fundamentals improving and we're certainly focused on returning to positive free-cash flow within quarterly positive free-cash flow within fiscal '24.

Q - Srini Pajjuri {BIO 5862807 <GO>}

Got it, but it doesn't look like you're assuming your inventory levels to decline on an absolute basis in the May quarter.

A - Mark Murphy {[BIO 16136048](#) <GO>}

Right. The sort of elevated inventory levels are in part contributing to this free-cash flow use that we have and we're certainly reducing CapEx and spend and other things to improve the situation and we do expect slight improvement in free-cash flow in the third quarter relative to the second, but it is still negative, significantly.

Q - Srinij Pajjuri {[BIO 5862807](#) <GO>}

Got it, thank you. And one question on the mechanics of the underutilization maybe for Manish. As we go through the next few quarters and as demand starts to recover, just curious if you expect pretty much all of the 100% of the underutilized capacity to come back or at what point some of the capacity becomes permanently impaired our move to the next node I guess? And then as you bring that back, is there any incremental CapEx that we should be aware of?

A - Manish Bhatia {[BIO 18045807](#) <GO>}

Srinij, I'm glad you asked that. It's actually definitely one of the considerations that we have when we think about underutilization is, as we implement these new technology nodes, we can utilize some of this idled equipment to help us be capital-efficient as we implement the next-generation nodes, right. So, there's always a balance that we're thinking about.

I wouldn't think of it in terms of any sort of impairment. I don't think that we're in that window. Our reuse node to node is very good. So, I wouldn't -- I don't think that idled equipment end-up getting impaired first to our consideration, but the ability to utilize some of the equipment to be a little bit more efficient as we transition to new nodes, because as we mentioned, new nodes could provide new product capabilities whether that higher-performance products or as we mentioned DDR5 capability or others. We're trying to make sure we balance the whole picture on Utilization, CapEx and demand for new technologies across our portfolio.

Q - Srinij Pajjuri {[BIO 5862807](#) <GO>}

I guess as you bring that capacity back online, is there any incremental CapEx, Manish, that we should be aware of in the short-term?

A - Manish Bhatia {[BIO 18045807](#) <GO>}

No, I mean not specifically for bringing online, but it depends on how -- we have multiple options of how we bring that capacity online in order to implement new technology nodes or to just replace the capacity that we had taken down before.

Q - Srinij Pajjuri {[BIO 5862807](#) <GO>}

Got it, thank you.

Operator

Thank you. This does conclude the question-and-answer session as well as today's program. Thank you, ladies and gentlemen, you may now disconnect. Good day.

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